

CLS LINAC Section Water Channel Test Procedure

1.7.15.1 Rev. 0

Date: 2011-11-02

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1.0 PURPOSE

This test procedure is used to test the water channels of the new LINAC acceleration sections to verify the channel conditions and to see if it meets with the design flow and pressure rating. The pressure test is to confirm and assure the water channels in these sections will perform function for which it was designed.

2.0 POLICY

The following standards are used as a guide to form this test procedure:

- CSA B51 – 03 Boiler, pressure vessel, and Pressure Piping Code;
- ASME B31.3-2006 – Process Piping Guide;
- The Boiler and Pressure Vessel Regulations (Saskatchewan - 2007);
- The Boiler and Pressure Vessel Act (Saskatchewan-1999);

3.0 SCOPE

This document includes a detailed leak test and a flow test procedure. The leak test is to confirm that the pressure rating at all water channels is still meeting with the design perimeter after shipping. The flow test is to confirm and assure the water channels in these sections will perform function with the flow rate and pressure drop for which they were designed. A sample of test record with required items to fill for acceptance is attached in Appendix A.

This document is for the testing of the CLSI new LINAC sections only.

4.0 DEFINITIONS

RF	Radio Frequency
LCW	Low Conductivity Water
PM	Water/gas pressure meter (in the drawing)
FLM	Water flow meter (in the drawing)
PCV	Pressure control valve/pressure regulator
PRV	Pressure relief valve
CF	Conflat

5.0 PROCESS

All water channels in each LINAC section are required test for leak and flow check. The leak test is to be performed by using Pneumatic leak test procedure. The flow test needs to be performed by using our current RF closed loop LCW system.

Nitrogen gas is chosen for the leak test for the reason of its non-flammable and nontoxic properties and the availability.

RF LCW water is chosen for the flow test for the following reasons:

- Same quality water as that is going to be used for this LINAC sections in the future;
- The closest operating pressure;
- Convenience of connections;

5.1 TEST PARAMETERS

- Leak test pressure: 150 PSI;
- Test temperature: room temperature between 20 °C and 40 °C;
- Leak test duration: Minimum 8 hours;
- Flow design criteria: 1.6 GPM @ 2 PSI pressure drop for each water channel;
- Flow test criteria: Total of 13 GPM flow rate through all eight parallel water channels at 50 psi water supply pressure;
- Pressure drop between water inlet and outlet: ≤ 2 PSI @ 13 GPM flow rate;

5.2 TEST SET UP SCHEMATIC

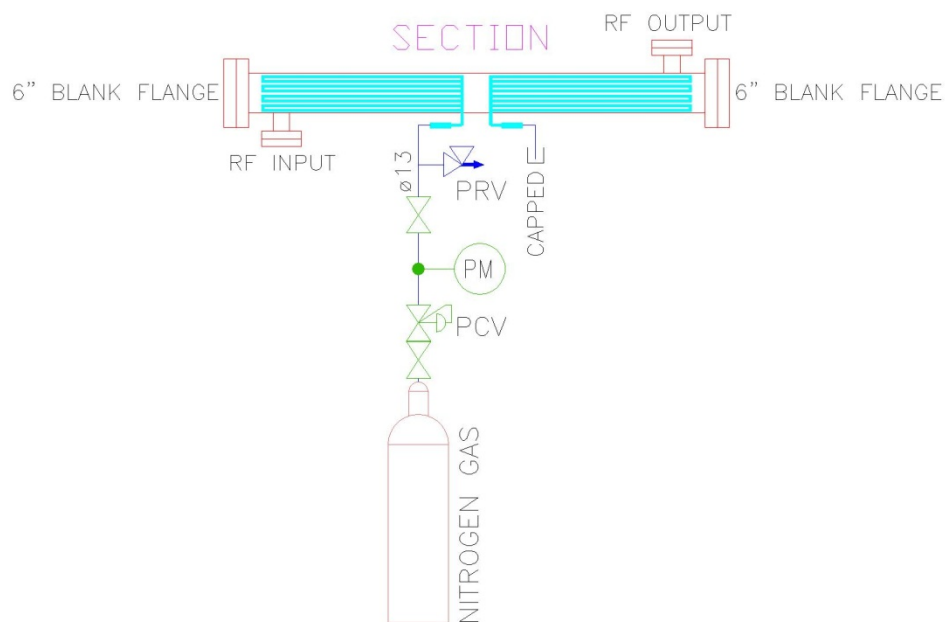


Figure 1. Leak Test Schematic

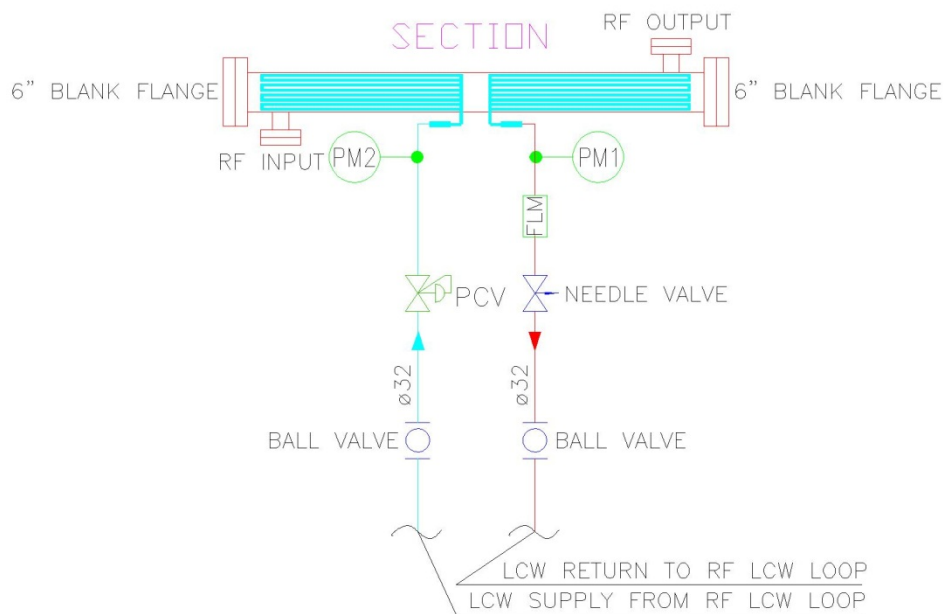


Figure 2. Flow Test Schematic

5.3 TEST EQUIPMENT AND PARTS LIST

Leak test:

- Two pieces of Skarpass male blank flanges for covering RF input and output;
- Two pieces of 6" blank CF flange;
- One nitrogen gas cylinder;
- One gas pressure gauge;
- One pressure control valve/pressure regulator;
- One pressure relief valve;
- One shut off valve;
- One end screw plug;
- 1/2" gas tubing ;
- Connection swag lock fittings;

Flow test:

- Two pieces of Skarpass male blank flanges for covering RF input and output;
- Two pieces of 6" blank CF flange;
- Two gas pressure gauges;
- One pressure control valve/pressure regulator;
- Two shut off ball valves;
- One flow control needle valve;
- One flow meter;
- 1 1/4" Water tubing ;
- Connection swag lock fitting;

5.4 TEST PROCEDURE

Leak test

- Seal two RF input and output ports with two Skarpass blank male flanges;
- Seal two 6" blank flanges at the two ends of section;
- Assembly the test system as the Leak Test Schematic shown in the Figure 1;
- Connect the test assembly to the section channel inlet and screw plug the section channel outlet connection port or the other way around;
- Pressure relief valve to set at 165 PSI;
- Pneumatic leak test: The pressure should be gradually increased until a gage pressure reaches 25 PSI. A preliminary check shall be made including examination of the joints and connections. Thereafter, the test pressure shall be gradually increased until it reaches the test pressure 150 PSI. Leave the system at the test pressure for minimum of 8 hours. The acceptance shall be made with no pressure decrease shown at the pressure gauge after the 8 hours;

Flow test (This test can only be performed for each section since the water channels are tied together with only one water inlet/outlet at each section)

- Seal two RF input and output ports with two Skarpass blank male flanges;
- Seal two 6" blank flanges at the two ends of section;
- Assembly the test system as the Flow Test Schematic shown in the Figure 2;
- Connect the test assembly to the section channel water inlet and outlet connection ports;
- Bring test LCW from RF LCW loop and set water supply pressure at pressure regulator to 50 psi (at pressure meter PM2);
- Open up all valves to see if flow can reach 13 GPM;
- Adjust needle valve and pressure regulator until flow meter reads exact 13 GPM. Read pressure gauge #1 & #2 to come up with pressure deference between. The maximum LCW supply pressure is 75 psi.

6.0 REFERENCE

- Final Manufacturing and Validation Report – RadiaBeam & IHEP;
- 1.8.32.1-Rev.0 - CLS LINPAC Acceleration Section Upgrade Specifications

7.0 ATTACHMENTS/FORMS

- 8.11.15.1 – Leak Test Record Sheet
- 8.11.15.2 – Flow Test Record Sheet

REVISION HISTORY

<i>Revision</i>	<i>Date</i>	<i>Description</i>	<i>Author</i>
A	2011-10-20	Original Draft	Linda Lin
0	2011-11-02	Issued for Use	Linda Lin



Leak Test Record Sheet

FORMNumber: **8.11.15.1** Rev. 0

Issued: 2011-11-02

TEST NAME:		PROJECT NUMBER:	
PROJECT NAME:			
TEST INFORMATION			
SYSTEM DISCRIPTION			
DESIGN TEMPERATURE: °C		DESIGN PRESSURE: PSI	
TEST METHOD: ☐ HYDROSTATIC ☐ PNEUMATIC			
TEST FLUID:		APPLICABLE CODE:	
TEST REQUIREMNTS			
REQUIRED TEST PRESSURE:		TEST FLUID TEMPERATURE:	
REQUIRED TEST DURATION:		AMBIENT TEMPERATURE:	
TEST RESULTS			
TEST DATE:	START TIME:	☐ AM ☐ PM	
	FINISH TIME:	☐ AM ☐ PM	
ACTUAL TEST DURATION:			
ACTUAL GAUGE PRESSURE (@ START):			
ACTUAL GAUGE PRESSURE (@ FINISH):			
REMARKS:			
TEST ACCEPTANCE:			
INSPECTOR:		DATE:	
CHECK/APPROVE:		DATE:	



Flow Test Record Sheet

FORM

Number: 8.11.15.2 Rev. 0
Issued: 2011-11-02

TEST NAME:		PROJECT NUMBER:	
PROJECT NAME:			
TEST INFORMATION			
SYSTEM DISCRIPTION			
DESIGN TEMPERATURE: °C		DESIGN PRESSURE: PSI	
DESIGN FLOW RATE FOR EACH CHANNEL:		NO. OF CHANNELS:	
DESIGN TOTAL FLOW RATE FOR EACH SECTION:			
DESIGN PRESSURE DROP:			
TEST METHOD: ☐ HYDROSTATIC ☐ PNEUMATIC			
TEST FLUID:		APPLICABLE CODE:	
TEST REQUIREMNTS			
REQUIRED TEST PRESSURE :		TEST FLUID TEMPERATURE:	
REQUIRED TEST PRESSURE (Max.):		AMBIENT TEMPERATURE:	
TEST RESULTS			
TEST DATE:	START TIME:	☐ AM ☐ PM	
	FINISH TIME:	☐ AM ☐ PM	
FLOW RATE AT 50 PSI LCW SUPPLY PRESSURE:			
WHEN FLOW RATE AT 13 GPM	PRESSURE AT PM#1:		
	PRESSURE AT PM#2:		
PRESSRUE DROP CROSS SECTION:			
REMARKS:			
TEST ACCEPTANCE:			
INSPECTOR:		DATE:	
CHECK/APPROVE:		DATE:	